

GESER DUGAROV, PhD

Software Engineer | Distributed Data Systems & AI Data Infrastructure

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profiles: [LinkedIn](#), [GitHub](#)
location: Novosibirsk, Russia | Open to relocation
languages: English (advanced), Russian (native)

SUMMARY

Software engineer building petabyte-scale lakehouse and AI data infrastructure at Huawei Cloud, with ownership from technical design through implementation, benchmarking, and production hardening. Author of [50 merged](#) Apache Hudi PRs and [three merged](#) Lance PRs; the Flink work reduced end-to-end TPC-H write time by up to 26% and inter-operator data transfer by 29–36%. Current work includes a two-tier distributed cache for Lance/Ray and a query-adaptive, incrementally updatable ANN index targeting tens of billions of vectors. Creator and operator of open-source [AI developer tooling](#), with 6,700+ agent runs recorded across 14 repositories.

TECHNICAL SKILLS

- **Languages:** Java, Python, SQL; Scala (working knowledge)
- **Distributed data and retrieval:** Apache Hudi, Apache Spark, Apache Flink, Lance, Ray, PostgreSQL, Parquet; lakehouse internals, distributed caching, vector search and ANN
- **Performance and reliability:** Serialization, checkpointing, partitioning/load balancing, profiling, memory/GC optimization, TPC-H benchmarking, monitoring
- **AI developer tooling:** Codex, Claude Code, multi-agent orchestration, human-in-the-loop (HITL) workflows, durable state/recovery, trajectory tracing, cost analytics
- **Engineering delivery:** Technical designs/RFCs, automated testing, CI/release engineering, Maven, Docker, Linux

WORK EXPERIENCE

May 2023 – Present **Software Engineer / Big Data Engineer,**
[Huawei](#)

Develop core lakehouse, distributed data, and AI retrieval infrastructure for Huawei Cloud; support teammates through code and RFC reviews and technical tutorials.

- Designed and implemented a two-tier distributed cache for Lance vector indices with an in-memory L1, persistent per-worker NVMe L2, and IVF-partition prewarming and routing; integrated it with Ray and benchmarked it on a real cluster.
- Developing a distributed, incrementally updatable, query-adaptive ANN index targeting tens of billions of vectors; authored [three merged](#) upstream Lance PRs that bounded IVF-training memory and fixed temporary-resource lifecycle issues, and helped resolve an out-of-memory problem in a three-billion-vector workload.
- Across [50 merged](#) Apache Hudi PRs, authored [RFC-84](#) and implemented its [Flink serialization redesign](#), cutting TPC-H write time by up to 26% and inter-operator data transfer by 29–36%; released in [Hudi 1.0.2](#).
- Four additional Flink optimizations ([\[1\]](#), [\[2\]](#), [\[3\]](#), and [\[4\]](#)) improved throughput by 10% and reduced GC overhead by 30%; released in [Hudi 1.0.1](#).

Huawei — continued

- Diagnosed serial buffer flushing during Flink checkpoints, contributing to a fix that improved checkpoint performance 7×; corrected writer load balancing and timestamp, CDC-merge, and cross-engine record-key defects.
- Simplified setup across 900+ Hudi configuration parameters; designed and implemented partition-level TTL for automated retention and storage-cost control, supported by automated tests and a 100,000-partition load test.
- Drove Hudi's Spark DataSource V2 migration; authored [RFC-98](#) and an initial Copy-on-Write [read path](#) to enable additional query pushdowns.

Feb 2022 – May 2023 **Software Engineer / ML Engineer,**
 Digital Research (computer vision startup)

- Designed the event-driven Python backend for a production truck-monitoring system processing ~20,000 images per day; the deployed system reduced fleet idle time by 12%. Also delivered customer reporting/data-visualization and internal monitoring interfaces.

OPEN-SOURCE ENGINEERING

[Agent Orchestrator](#) – Creator & Maintainer | Apr 2026 – Present

- Designed and shipped a Python issue-to-PR platform that orchestrates isolated implementation, automated validation, independent review, and human-controlled merge using Codex and Claude Code.
- Built durable GitHub-persisted workflow state, pause/resume and recovery, credential isolation, audit/trajectory logging, and PostgreSQL cost analytics. Shipped 12 releases through v0.11.0 and recorded 6,700+ agent runs across 14 repositories, including 2,000+ independent review-agent runs.

EDUCATION

PhD, Geophysics, Trofimuk Institute of Petroleum Geology and Geophysics SB RAS

MSc, Computational and Applied Mathematics, Novosibirsk State University